
The consequences of climate change in Atlantic Forest birds and mammals

A Data Management Plan created using DMP Assistant

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Project abstract:

Changes to body size and shape have been identified as potential adaptive responses to climate change, but how widespread these responses are remains controversial. To contribute to this debate, we will use data from two datasets of morphological traits of birds and mammals from the ecological rich and environmentally threatened Atlantic Forest in Brazil. Overall, our findings will allow us to test whether morphological adaptation is a widespread response to climate change in birds and mammals. We will also use datasets of climatic variables as our explanatory variables. Finally, we will use datasets of arthropod abundance over time as another one of our ecological explanatory variables.

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Responsibilities and Resources

Who will be responsible for data management? Will the Principal Investigator (PI) hold all responsibility during and beyond the project, or will this be divided among a team or partner organizations?

Data management will be the responsibility of the Principal Investigator (PI), Eduardo S.A. Santos, and will be shared with team members. The PI will be the person with most responsibility over data management in this project.

In the event that the PI leaves the project, who will replace them? Who will take temporary responsibility until a new PI takes over?

In case the PI leaves the project, the person that will take temporary responsibility over the data is Dr. Gustavo Burin.

List all expected resources for data management required to complete your project. What hardware, software and human resources will you need? What is your estimated budget?

Most of the original data that will be used in this project is available publicly from published sources. As such, we do not need any additional costs to store original raw data that will be used. For the duration of the active phases of this project, data being used can be stored locally at the University of Alberta's RDSS system. Secondly, data can be backed-up to cloud solutions available for team members, such as Google Drive storage. All resources that we will need to complete this project with regards to data management, are already available through the PI's research unit.

Data Collection

What types of data will you collect, create, link to, acquire and/or record?

The main data that will be used in this project comes from publicly available previously published data from the Atlantic Bird Traits (<https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecy.2647>) and Atlantic Mammal Traits (<https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.2106>) datasets. These datasets are available from online repositories as *.csv files, or via an R package as objects from the library.

We will also use the Elton traits dataset to extract information about functional/foraging attributes of birds and mammals. These datasets are also publicly available as *.txt files (<https://doi.org/10.6084/m9.figshare.c.3306933.v1>).

Data on the phylogenetic relationship of the birds and mammals will also be collected from published studies (**Insert sources of phylogenetic information here**). These files are usually available as *.nexus

Data on arthropod abundance will be collected from the PREDICTS dataset (<https://www.nhm.ac.uk/our-science/research/projects/predicts.html>) and (https://data.nhm.ac.uk/dataset/bii-bte?_gl=1*1ynp1pi*_*ga*NzQzNyYlNjYzLjE3NjA2NDU4NzI_*_ga_PYMKGK73C4czE3NjA2NDU4NzEkbzEkZkZkdE3NjA2NDYxMDUkaYwJGwwJGgw). *PREDICTS data is available as .csv.*

Answer the following regarding file formats:

- What file formats do you expect to collect (e.g. .doc, .csv, .jpg, .mov)?
- Are these file formats easy to share with other researchers from different disciplines?
- In the event that one of your chosen file formats becomes obsolete (or is no longer supported) how will you ensure access to the research data?
- Does your data need to be copied to a new media or cloud platform, or converted to a different file format when you store or publish your datasets?

1. We expect to collect files as .csv, .txt, .nexus, .rda, and .rds;
2. These file formats are easy to share with other researchers;
3. All the file types we will collect and use for our data in this project are easily accessible with cross-platform and free source software, such as text editors, and free statistical analysis software; and
4. When we store or publish our datasets we will need to copy it to cloud platforms so that it is available to the public. We will use data storage repositories for that purpose.

Answer the following regarding naming conventions:

- How will you structure, name and version-control your files to help someone outside your research team understand how your data are organized?
- Describe your ideal workflow for file sharing between research team members step-by-step.
- What tools or strategies will you use to document your workflow as it evolves during the course of the project?

1. Naming conventions will be used for our files so that files are understandable by the research team and people from outside our research group. Files will use standard convention, including dates, data state (raw, accurate...). Structure will help to explain if data comes from origins (sources) or if it was generated by the study (results);
2. Our workflow for data usage and sharing will be described in our github repository for this research project. All members of the team have access to this github repository; and
3. We will use github as our tool for version control and to document the workflows of this project.

Documentation and Metadata

What support material and documentation (e.g. ReadMe) will your team members and future researchers need in order to navigate and reuse your data without ambiguity?

We will create a ReadMe file to instruct users on understanding our data collection and processing methods, naming conventions and file structure. Our documentation will include information about the study, data-level descriptions, and any other contextual information required to make the data usable by other researchers. We will also document: research methodology used, variable definitions, vocabularies, classification systems, units of measurement, assumptions made, format and file type of the data, a description of the data capture and collection methods, explanation of data coding and analysis performed (including syntax files).

How will you undertake documentation of data collection, processing and analysis, within your workflow to create consistent support material? Who will be responsible for this task?

Documentation of data collection, processing and analysis will be recorded in markdown files within our github repository, so that it is reproducible and shareable. All team members will be responsible for this task.

Do you plan to use a metadata standard? What specific schema might you use?

We have not thought about using a metadata standard, but it will surely be considered and made available, if possible.

How will you make sure that a) your primary data collection methods are documented with transparency and b) your secondary data sources (i.e., data you did not collect yourself) — are easily identified and cited?

Secondary data sources will be easily identified throughout our protocol and manuscript, with appropriate citation and path to the archived version of the data. Our primary data (generated by our study) will be deposited in repositories with static DOI numbers and paths will be made available in our protocols and manuscript.

Storage and Backup

List your anticipated storage needs (e.g., hard drives, cloud storage, shared drives). List how long you intend to use each type and what capacities you may require.

Our datasets (both original sources and resulting datasets) will be stored locally in the PI and team members harddrives, via version control repositories. Also, data will be store in the cloud (University of Alberta Google Drive) and the RDSS storage shared drive service of the University of Alberta. All necessary storage needs are already met at the beginning of this project.

What is your anticipated backup and storage schedule? How often will you save your data, in what formats, and where?

Data will be version controlled and backed-up. Back up will take place in cloud storage and University of Alberta's shared drive RDSS, as well as across team member's working computers.

Keeping ethics protocol review requirements in mind, what is your intended storage timeframe for each type of data (raw, processed, clean, final) within your team? Will you also store software code or metadata?

The timeframe of storage for all the data types of our project will run the length of the active project phase. After the work is finalized and published, we will store backup copies of the datasets in cloud storage services of team members through version control.

Sharing, Reuse, and Preservation

How will your data (both raw and cleaned) be made accessible beyond the scope of the project and by researchers outside your team?

We plan to make our data accessible to other by depositing our data in a repository such as Zenodo or Dryad.

Is digital preservation a component of your project and do you need to plan for long-term archiving and preservation?

The data generated from our study will need to be archived for long term access for reuse. We will ensure that our data will be made available in file formats that are long lasting and of efficient storage formats.

What data will you be sharing publicly and in what form (e.g. raw, processed, analyzed, final)?

We will share raw, processed, analyzed and final datasets from our study.

Have you considered what type of end-user license to include with your data?

Our datasets will be licensed as CC-BY, which will allow others to copy, distribute, remix and build upon our work—as long as they credit and cite our work.

What tools and strategies will you take to promote your research? How will you let the research community and the public know that your data exists and is ready to be reused?

Our data will be published in a repository and will have a persistent DOI identifier. This dataset will be associated with the final version of our published manuscript.

Ethics and Legal Compliance

Are there institutional, governmental or legal policies that you need to comply with in regards to your data standards?

Data being used in our study comes from secondary sources (it has already been published and made publicly available in repositories). All data used in our study comes from non-human organisms. Therefore, our data complies to policies.

Will you encounter protected or personally-identifiable information in your research? If so, how will you make sure it stays secure and is accessed by approved team members only?

We will not encounter protected or personally-identifiable information in our research.

Before publishing or otherwise sharing a dataset are you required to obscure identifiable data (name, gender, date of birth, etc), in accordance with your jurisdiction's laws, or your ethics protocol? Are there any time restrictions for when data can be publicly accessible?

This is not applicable, as we will not work with identifiable data.

Planned Research Outputs

Dataset - "Dataset for The consequences of climate change in Atlantic Forest birds and mammals"

Planned research output details

Title	Type	Anticipated release date	Initial access level	Intended repository(ies)	Anticipated file size	License	Metadata standard(s)	May contain sensitive data?	May contain PII?
Dataset for The consequences of climate change in ...	Dataset	Unspecified	Open	Zenodo		None specified	None specified	No	No